Default Question Block

Thank you for participating in this study – we are truly grateful.

This project is investigating attitudes and awareness of artificial intelligence (AI) in medical genetics.

We will first ask a few general questions including whether you are willing to participate and general demographics related to your practice. We anticipate that participation will take than 20 minutes (most participants take about 10 minutes to complete).

We will then ask you to answer a series of multiple-choice questions on your familiarity with AI and awareness of the current uses of AI in clinical genetics. Once you have finished this portion of the survey, you will be asked to complete ranked responses to a series of opinion statements regarding the use of AI in the clinical genetics workspace.

You may take this survey on your smart device or your desktop/laptop computer. Once you answer a question, you will not be able to go back.

No identifiers will be collected as part of the survey. All individual results will be maintained in an anonymous format. You will not receive your individual results. We plan to describe/publish only deidentified results.

Your participation is voluntary and will not be compensated. By continuing with the survey, you agree to take part.

If you have questions, encounter any difficulties, or change your mind about participation, please contact the lead investigator Amanda Berkstresser at aberkstresser2201@baypath.edu or 816.872.4506 or Daniela Iacoboni at diacoboni@baypath.edu.

- Continue to survey
- View full consent, then continue to survey

Consent to be Part of a Research Study

Title of the Project: Attitudes and Awareness of Artificial Intelligence in Clinical Genetics
Principal Investigator: Amanda Berkstresser
University Department: Genetic Counseling
Faculty Advisor: Daniela Iacoboni, LGC, Bay Path University

Invitation to be Part of a Research Study

You are invited to participate in a research study. To participate, you must be a board certified/board-eligible genetics professional (MD, DO, GC, NP, PA, or RN) or a genetics trainee (student, resident, or fellow). All participants must be at least 18 years of age with proficiency in the English language. Taking part in this research project is voluntary.

Important Information about the Research Study

Things you should know:

- This project is investigating attitudes and awareness of artificial intelligence (AI) in medical genetics. If you choose to participate, you will be directed to an online survey and asked to complete a series of multiple-choice and ranked responses. We anticipate your participation to take 20 minutes or less.
- There are no anticipated risks or discomforts from this research.
- While your participation has no immediate benefit, your responses can help expand upon available information on this topic.
- Taking part in this research project is voluntary. You don’t have to participate and you can stop at any time.

Please take time to read this entire form and ask questions before deciding whether to participate in this research project.

What is the study about and why are we doing it?

This research seeks to better understand attitudes and awareness of the use of AI in medical genetics, with the intent of becoming better informed regarding the development and integration of future applications.

What will happen if you take part in this study?

If you agree to take part in this study, you will be directed to an online survey link where you will first be asked to answer general demographics questions related to your genetics practice. We will then ask you to answer a series of multiple-choice questions on your familiarity with AI and awareness of the current uses of AI in clinical genetics. You will conclude the survey by completing ranked responses to a series of opinion statements. It is expected to take approximately 20 minutes to complete.

How could you benefit from this study?

There is no immediate benefit to you for your participation. Your responses may be used to help guide the development of future AI innovations.

What risks might result from being in this study?

There are no identifiable risks from participating in this research.

How will we protect your information?

We will protect the confidentiality of your research records by not collecting any identifying information. Your responses are anonymous. Data will be stored via Qualtrics (the survey platform) and may be transferred to Excel for statistical analysis.

The results of this study will be presented to medical genetics professionals and students and may be published in a peer-reviewed professional journal. To protect your privacy, no information that could directly identify you will be included.

What will happen to the information we collect about you after the study is over?

Your survey responses will be kept for reference in understanding attitudes and awareness of AI in the clinical genetics workspace to guide the development of future innovations. Your name and other information that can directly identify you will not be recorded.

We will not share your research data with other investigators.

How will we compensate you for being part of the study?

There is no compensation for your participation.

Your Participation in this Study is Voluntary

It is totally up to you to decide to be in this research study. Participating in this study is voluntary. Even if you decide to be part of the study now, you may change your mind and stop at any time. You do not have to answer any questions you do not want to answer. If you decide to withdraw before this study is completed, only the questions you responded to will be recorded.

Contact Information for the Study Team and Questions about the Research

If you have questions about this research, you may contact Amanda Berkstresser (PI) at **816-872-4506** or **aberkstresser2201@baypath.edu,** or Daniela Iacoboni at **diacoboni@baypath.edu**.

Contact Information for Questions about Your Rights as a Research Participant

If you have questions about your rights as a research participant, or wish to obtain information, ask questions, or discuss any concerns about this study with someone other than the researchers, please contact the following:

Bay Path University
Office of Academic Affairs
588 Longmeadow Street
Longmeadow, MA 01106
Phone: (413) 565-1000

Your Consent

Before clicking on the link below to start taking the survey, please make sure you understand what the study is about. By clicking on the link below, you are agreeing to be in this study and are also confirming you are 18 years of age or older. You can print a copy of this document for your records. If you have any questions about the study later, you can contact the study team using the information provided above.

- Continue to survey

Demographic Questions

I am

- a board certified or board eligible genetics professional (MD, DO, GC, RN, NP, or PA)
- a genetics trainee (student, resident, or fellow)

Please select your medical profession.

- Medical Geneticist
- Genetic Counselor
- Physician Assistant
- Nurse Practitioner
- Registered Nurse

For how long (since you became board eligible) have you been practicing?

- < 1 year
- 1 to 5 years
- 5 to 10 years
- > 10 years

How would you describe your current position?

- Primarily clinical
- Primarily research
- Approximately equal clinical and research
- Administrative
- Sales/Marketing

- Clinical laboratory
- Education/Teaching
- Other

Please describe your current position.

How would you describe your institution?

- Community based hospital
- Academic medical or research center
- Molecular diagnostic company
- Telehealth company
- Other

Please describe your institution.

Please select your program.

- medical genetics resident
- medical genetics fellow
- genetic counseling student
- other

Please write in the name of your training program.

What year are you in your genetics training program?

- Year 1
- Year 2
- Year 3
- Year 4 or more

What is your age range?

- Under 30 years old
- 30 to 45 years old
- 46 to 60 years old
- Over 60 years old

Please enter your state (if in the US) and/or country.

Familiarity with AI

What is your current level of knowledge regarding the use of AI in clinical genetics?

- Little to no knowledge of AI in clinical genetics
- Intermediate knowledge of AI in clinical genetics
- Extensive knowledge of AI in clinical genetics

How often do you use AI in your clinical practice?

- Daily
- Weekly
- Monthly
- Rarely
- I do not use AI in clinical practice

How often do you think AI is used in the clinical setting?

- Daily
- Weekly
- Monthly
- Rarely
- I do not use AI in clinical practice

Approximately how often do you think a typical genetics clinician utilizes AI in the clinical setting?

- Daily
- Weekly
- Monthly
- Rarely
- They do not use AI in clinical practice

Are clinical AI tools accessible to you?

- Not accessible
- Most tools not accessible
- Some tools acessible
- Accessible

How user-friendly do you find clinical AI tools?

- Very difficult to use
- Somewhat difficult to use
- Somewhat easy to use
- Very easy to use

How beneficial do you think AI tools are in the current practice of clinical genetics?

- Not beneficial
- Beneficial
- Highly beneficial

Have you ever attended a workshop or seminar that addressed the utilization of AI in healthcare?

- yes
- no

Did you receive training in AI during graduate/professional school?

- There was no training offered regarding AI.
- I received a small amount training regarding AI.
- I received a moderate amount of training regarding AI.
- I received extensive training regarding AI.

Current uses of AI in clinical genetics

You will next see 6 examples of the uses of AI in clinical genetics. Please answer the questions following each short passage.

Differential Diagnosis:
Technology has long been integrated into and utilized for the process of generating differential diagnoses, beginning with data retrieval approaches like SYNDROC and Phenomizer, then moving to tools that employ computer vision for facial gestalt image analysis, like Face2Gene, which uses a type of AI called deep learning.

Have you used AI-based facial diagnostics, such as Face2Gene, as a differential diagnosis tool?

- Yes
- No
- Unsure

Do you think you will use AI-based facial diagnostics, such as Face2Gene, as a differential diagnosis tool when you become a practicing GC?

- Yes
- No
- Unsure

Prior to this survey, were you aware that Face2Gene is an AI-based tool?

- Yes
- No

Clinical Genomic Testing:
AI tools are routinely employed in clinical genetic/genomic testing. Phenotypic information can be quickly gathered from clinical notes using a type of AI called natural language processing (NLP). DNA is typically sequenced and at least in part interpreted by AI technology. For example, Illumina NGS DNA sequencers use the DRAGEN Pipeline, which utilizes AI. Lastly, many molecular diagnostic companies are using AI tools to do things like predict the functional impact for variant interpretation.

Have you used molecular diagnostic companies that employ AI in their genomic testing process?

- Yes
- No
- Unsure

Do you think you will use molecular diagnostic companies that employ AI in their genomic testing process when you become a practicing GC?

- Yes
- No
- Unsure

Does your lab utilize AI in the genetic/genomic testing process?

- Yes
- No
- Unsure

Prior to this survey, were you aware AI was used in clinical genomic testing?

- Yes
- No

Imaging Diagnostics:
AI based tools can be used to analyze MRI, CT, or ultrasound, as well as other types of medical images, to assist in diagnosis. AI tools have been shown to potentially help with the detection of abnormal cardiac function (for MRI and echocardiograms), brain bleeds and anomalies (for CT scans), cancers (for X-rays and MRI), as well as to predict breast cancer risk based on a seemingly normal mammogram. For example, AI-driven analysis of normal mammograms outperformed standard breast cancer risk models for predicting the five-year risk of breast cancer [PMID: 37278627].

Have you reviewed patient imaging results that were interpreted by AI?

- Yes
- No
- Unsure

Do you think you will review patient imaging results that were interpreted by AI when you become a practicing GC?

- Yes
- No
- Unsure

Have you reviewed patient imaging results that were interpreted by AI?

- Yes
- No

Chatbots:
Recent chatbots like ChatGPT, Llama2, and Bard are built based on the principle of large language model. That is, by learning the correlations of words from very large text databases, these chatbots can predict the next probable words based on the user’s query. For example, there are instances for which ChatGPT can diagnose a patient [PMID: 37578798]. However, ChatGPT is not built specifically for medical purposes; whereas Invitae’s Genetic Information Assistant (Gia) is one such example. Gia is an AI-based, HIPAA-compliant chatbot, that provides pre- and post-test education. Additionally, many

hospitals and medical groups have created their own chatbots to assist in genetic testing education and obtain informed consent.

Have your patients used chatbots, like Gia, for genetic testing education and/or consenting?

- Yes
- No
- Unsure

Do you think your patients will use chatbots, like Gia, for genetic testing education and/or consenting when you become a practicing GC?

- Yes
- No
- Unsure

Prior to this survey, were you aware medical based chatbots like Gia use a type of AI?

- Yes
- No

Have you used ChatGPT or other LLM to assist in the diagnosis or management of a patient?

- Yes
- No
- Unsure

Do you think you will use ChatGPT or other LLM to assist in the diagnosis or management of a patient when you become a practicing GC?

- Yes
- No
- Unsure

Prior to this survey, were you aware that ChatGPT and LLMs are AI-based tools?

- Yes
- No

Administrative Tasks:
In a workload-heavy field, AI can be applied to administrative tasks. AI tools through natural language processing can complete intake forms, construct family pedigrees, and analyze electronic health records. This allows clinicians to focus on specific subsets of patients or certain aspects of an individual's medical records to facilitate efficient diagnosis and treatment.

Do you use AI-prepared documents such as pedigrees and medical summaries in your practice?

- Yes
- No
- Unsure

Do you think you will use AI-prepared documents such as pedigrees and medical summaries in your practice when you become a practicing GC?

- Yes
- No
- Unsure

Prior to this survey, were you aware AI-based tools could assist in constructing medical summaries and pedigrees?

- Yes
- No

Risk Assessment:
Risk assessment platforms are available to aid in risk calculation. Standard risk assessment has involved risk calculator tools like the Tyler-Cuzick model (breast cancer)

and web-based platforms such as CancerGeneConnect, MeTree, and Family Healthware, which can combine pedigree construction with risk assessment to create an individualized risk model. The most recent technology uses AI to combine polygenic risk score with clinical information to provide a more accurate risk assessment. Examples include risk scores for breast cancer (BRECARDA) and cardiovascular disease (NeuralCVD and DeepCausalAI).

Do you use AI-based risk assessment tools in your practice?

- Yes
- No
- Unsure

Do you think you will use AI-based risk assessment tools in your practice when you become a practicing GC?

- Yes
- No
- Unsure

Prior to this survey, were you aware that some risk assessment tools use AI?

- Yes
- No

Follow-up evaluation of familiarity with AI

Now that we have described some types of AI used in clinical genetics, we will re-ask a few questions to allow you to update your answers.

How often do you use AI in your clinical practice?

- Daily
- Weekly
- Monthly
- Rarely

- I do not use AI in clinical practice

How often do you think AI is used in clinical setting?

- Daily
- Weekly
- Monthly
- Rarely
- I do not use AI in clinical practice

Approximately how often do you think your colleagues utilize AI in your clinical settings?

- Daily
- Weekly
- Monthly
- Rarely
- They do not use AI in clinical practice

Are clinical AI tools accessible to you?

- Not accessible
- Most tools not accessible
- Some tools acessible
- Accessible

How beneficial do you think AI tools are in the current practice of clinical genetics?

- Not beneficial
- Beneficial
- Highly beneficial

Likert Scale Questions

I am comfortable using AI in my clinical genetics practice.

Strongly disagree

Disagree Agree

Strongly agree

AI can be safely used in the handling of confidential data.

Strongly disagree

Disagree Agree

Strongly agree

AI provides reliable outputs.

Strongly disagree

Disagree Agree

Strongly agree

Using AI as an aid is outside my scope of training/practice.

Strongly disagree

Disagree Agree

Strongly agree

AI is inappropriate for use as a diagnostic support tool.

Strongly disagree

Disagree Agree

Strongly agree

I have felt forced to utilize AI in my practice.

Strongly disagree

Disagree Agree

Strongly agree

I am concerned AI will not be used responsibly in clinical genetics.

Strongly disagree

Disagree Agree

Strongly agree

AI is not ready to be used in clinical genetics.

Strongly disagree

Disagree Agree

Strongly agree

My patients would be comfortable with the use of AI in clinical genetics.

Strongly disagree

Disagree Agree

Strongly agree

I am comfortable with my patients using AI to obtain general genetics information.

Strongly disagree

Disagree Agree

Strongly agree

I am comfortable with my patients using AI to undergo pretest genetic counseling.

Strongly disagree

Disagree Agree

Strongly agree

Clinical genetics professionals are overburdened.

Strongly disagree

Disagree Agree

Strongly agree

AI is an appropriate solution to relieving workload.

Strongly disagree

Disagree Agree

Strongly agree

I am willing to delegate work to other people.

Strongly disagree

Disagree Agree

Strongly agree

I am willing to delegate work to AI.

Strongly disagree

Disagree Agree

Strongly agree

There should be further curriculum and education about the use of AI in clinical care.

Strongly disagree

Disagree Agree

Strongly agree

I currently lack sufficient knowledge about AI in medical genetics.

Strongly disagree

Disagree Agree

Strongly agree

If offered a class about AI in clinical genetics, I would take it.

Strongly disagree

Disagree Agree

Strongly agree

The development of new medical genetics AI tools requires input from clinical genetics professionals.

Strongly disagree

Disagree Agree

Strongly agree

There needs to be further regulations for AI before I incorporate it into my genetics practice.

Strongly disagree

Disagree Agree

Strongly agree